REAL TIME TRAFFIC DENSITY ESTIMATION AND MANAGEMENT USING ML

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***Abstract*— One of the greatest problems in the city is traffic congestion, which causes a longer journey period, higher usage of fuel and pollution of the environment. The conventional system of traffic management is based on set timers or manual control that do not adjust to the traffic real-time conditions. This paper offers a machine learning and computer vision-based traffic density estimation and management system in real-time. The system will use CCTV video feeds to track the traffic and use the YOLOv11 object detection model to detect vehicles accurately and effectively in complex and dense traffic conditions. Owing to the identified number of vehicles, traffic density is approximated and implemented to dynamically change signal timing in a rule-based optimization strategy. The system is developed in Python, OpenCV, and Flask and a real-time dashboard to monitor the traffic condition and states of the signals. Experiments show that the suggested system can deliver effective and reliable traffic control to be used in smart cities.**

I. INTRODUCTION

Traffic jam is among the most urgent problems of contemporary urban setting caused by the intensive urbanization and the constant rise in the number of automobiles. It causes delays in the travelling time, wastage of fuel, environmental pollution and commuter stress. Traffic control has thus been a major part of the sustainable urban development and intelligent transportation systems (ITS). The conventional traffic surveillance tools are based on the infrastructural means like an inductive loop sensor, infrared detector, and human observation. Despite all these benefits of these methods, they have a number of limitations such as high cost of installation and maintenance, limited coverage, not scalable, and not able to dynamically respond to the real time traffic conditions.

Vision-based traffic monitoring systems have become an alternative to camera-based ones, with the development of machine learning (ML) and computer vision (CV), being more flexible and cost-effective. The presence of CCTV cameras across most cities allows one to monitor traffic 24 hours without the use of new equipment. Object detection models that are built using deep learning may process such video feeds to detect and classify vehicles in real time, even in complex and dense traffic conditions.

The YOLO (You Only Look Once) family of models has one of the biggest followings because it works fast and is rather accurate in object detection tasks. Nevertheless, the use of recent developments like YOLOv11, which has several improvements in detecting performance, including better feature extraction and multi-scale detection, can be applicable under urban traffic conditions where vehicles can be seen as either small, distant, or partly covered. YOLOv11 has been used in this work to identify vehicles in real-time video frames and deliver quality information to traffic analysis.

The density of the traffic based on the identified vehicles is estimated as the number of vehicles in a specified area of interest. The proposed system does not utilize complex tracking-based algorithms as do approaches based on density estimation; rather, it employs a real-time detection-based density estimation that minimizes the computational load on the computer, yet the proposed approach is practical and achieves reasonable accuracy. With the estimated density, traffic signal timings are then dynamically adjusted by a rule-based optimization mechanism which is better than the traditional fixed-timer systems in improving traffic flow.

Moreover, the system has a real-time dashboard that demonstrates the traffic status, identified vehicles, and signal lights, allowing to monitor and make decisions. In general, the smart traffic management approach suggested is scalable, cost-efficient, and automated, and thus it can be used in the smart city.

II. LITRATURE SURVEY

Traffic jam has been a significant issue in all urban transportation networks and this has necessitated massive research on traffic monitor, traffic density estimation, and intelligent traffic control. Initial methods were relying much on human surveys with a fixed structure like inductive loop detectors, radar sensors, and infrared counters. Although these systems offered rudimentary traffic information, they were characterized by expensive costs of installation and maintenance, sparse coverage and lack of flexibility to dynamism of the actual traffic conditions. These shortcomings prompted scientists to investigate data-driven and vision-driven methods of scalable and real-time traffic analysis.

2.1 Vision-Based Vehicle Detection

Vision-based traffic systems that were first introduced utilized the classical image processing methods like background subtraction, edge-detection, and optical flow. Though they were computationally efficient, these techniques were sensitive to environmental variations, such as changes in lighting, shadows, occlusions and weather conditions, which made them very poor in the real world. Detection accuracy and strength were greatly enhanced with the introduction of deep learning. Object detection models like Faster R-CNN, SSD and YOLO transformed object detection by facilitating end-to-end learning. Faster R-CNN had high accuracy and high computational latency since it is a two-stage architecture that is not applicable in real-time applications. SSD enhanced speed yet it had trouble in detecting small and distant vehicles which are prevalent in traffic surveillance situations. YOLO (You Only Look Once) family of models became popular because of single-stage architecture which provides a trade-off between speed and accuracy. The YOLOv3 introduced multi-scale detection and the YOLOv4 improved it by introducing better architecture and optimization of training. More modern models, including YOLOv8, also maximized faster inference and adaptability. Nevertheless, difficulties are still experienced in identifying small, distant, and congested vehicles in tricky traffic situations. These drawbacks encourage the use of more sophisticated models including YOLOv11 that enhances the representation of features and detection capabilities in heavy traffic conditions.

2.2 Traffic Density Estimation Techniques

In addition to the detection, traffic density estimation is necessary in congestion analysis and traffic control. A number of methods approximate density by counting the number of vehicles observed in a single frame, whereas density-map-based convolutional neural networks approximate the traffic flow without specifically detecting individual vehicles. Methods using density-maps do not need as much annotation, but are not capable of classifying individual vehicles, making them less useful in finer-grained traffic analysis. They have been extensively used in detection-based strategies uses in conjunction with Region of Interest (ROI) analysis in which certain areas of the road are examined to determine the congestion level by the number of cars. Nevertheless, most of the current systems have complicated tracking systems to sustain vehicle identity among frames that add computational costs. Contrarily, simpler detection-based counting approaches are more favorable in real-time systems since they are accurate enough and their computation is computationally efficient enough to implement in practice.

2.3 Traffic Signal Control Systems

The conventional traffic signal systems are majorly fixed-timed or manually operated that fails to respond to real-time traffic. In order to tackle this weakness, a number of intelligent traffic control strategies have been suggested. Signal timing optimization has been optimized using machine learning-based models, such as regression, decision trees and ensemble algorithms, such as Random Forest and XGBoost. LSTM networks and GRU networks are examples of deep learning models that have been demonstrated to be effective to predict traffic patterns and congestion due to traffic. Also, the adaptive signal control has been studied using reinforcement learning and graph models. Their effectiveness in spite of these methods usually demand vast amounts of past information, substantial computing assets, and complicated integration of systems, which are hard to execute in real-time, camera-based settings. Subsequently, simpler rules-based optimization approaches and real-time detection still present a viable and effective resolution.

2.4 Identified Research Gaps

According to the available literature, there are a number of major gaps that could be identified in the existing traffic management systems. Most of these methods have a difficult time detecting small and far-away vehicles and particularly in dense and complicated traffic conditions. Moreover, many systems have sensor-based infrastructure that constrains scalability as well as raises the cost of installing and maintaining the systems. Complex tracking algorithms are used in some of the techniques and present a high computational cost, causing real-time implementation to be difficult. Also, little emphasis is placed on applying vehicle detection to real-life implementation of traffic signal optimization. Lastly, a large number of sophisticated models cannot be easily deployed by showing high complexity and resource needs and limiting their deployment in real-time traffic settings.

2.5 Motivation of the Proposed Work

To overcome these issues, the suggested system employs YOLOv11 to achieve effective and precise vehicle recognition on a real-time traffic setup. The system uses a detection-based method in estimating traffic density, as opposed to using complicated tracking systems, thereby making the computation process less complex, but preserving practical accuracy. Traffic lights are dynamically regulated with an estimate of the density based on the rule-based optimization algorithm and the traffic conditions can be correctly adjusted in real-time. As well, a visualization system is represented via a dashboard with real-time monitoring and analytics, which aids in making a decision. The solution is a cost-effective, scalable, and real-time intelligent traffic management approach, which is appropriate in terms of implementation in the contemporary smart city setting.

III. PROPOSED METHODOLOGY

This paper proposes real time estimates of traffic density and traffic management based on computer vision and machine learning algorithms. The system employs CCTV video feeds to check the traffic conditions and uses YOLOv11 to detect vehicles and ambulances. The traffic density is estimated based on the number of vehicles detected and utilized in the dynamic traffic signal control.

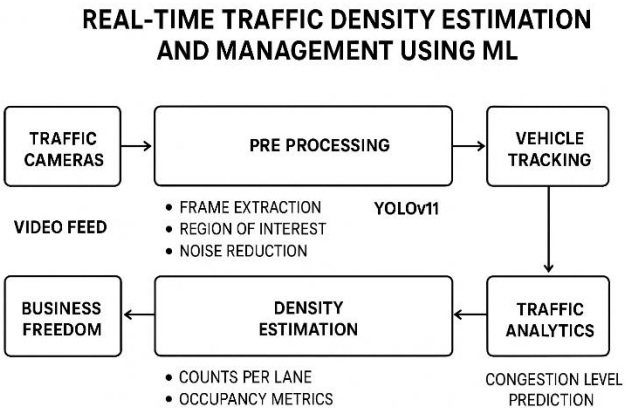


FIG.1: System Architecture Diagram

The proposed methodology is divided into multiple functional modules, each responsible for a specific stage of the system pipeline.

A. Data Collection and Preparation

The traffic video data is recorded based on the real-time feeds of CCTV cameras and publicly accessible data. The dataset comprises different traffic scenarios of different vehicle densities, changes in lighting conditions, weather changes and different road environments. This heterogeneity makes the system work reliably in the real-life situation. The gathered data is applicable mainly in testing and confirming the performance of the system as opposed to training a sophisticated model pipeline.

B. Video Input and Pre-processing Module

Live CCTV video feeds or uploaded traffic video files are used to receive the video input. With the help of the OpenCV library, a video stream is disaggregated into single frames, which are processed in real-time. All the frames are pre-processed by the following stages: Resizing to constant resolution, Noise reduction to eliminate environmental disruption, Normalization to sustain a constant pixel intensity. Furthermore, an ROI is introduced, which will only compute on areas that are of interest and will enhance the features of the computation and accuracy of the detection.

C. Object Detection Module (YOLOv11)

The frames that have been pre-processed are fed into the YOLOv11 object detector model. YOLOv11 real time detects cars, buses, trucks, and motorcycles, and ambulances. The model produces one: Bounding box co-ordinates, Class label Only detections with a specified confidence threshold are accepted to minimize the false positives. YOLOv11 is chosen because it is high-speed and capable of identifying several objects in a traffic jam effectively.

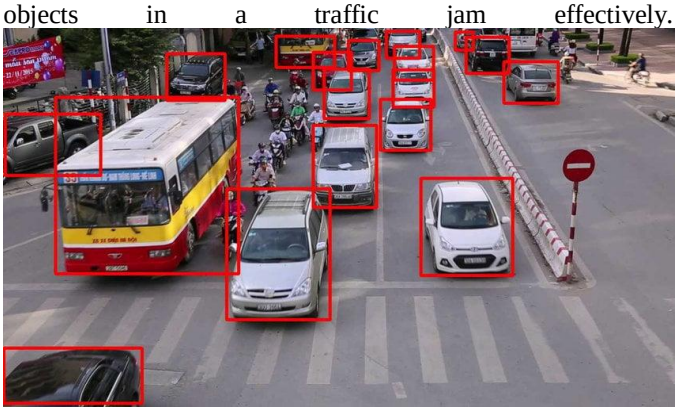


FIG.2: Vehicles Detection

D. Traffic Density Calculation Module

The density of the traffic is calculated according to the estimation of the number of vehicles that are detected in the area of interest per frame.

Let: N= Number of detected vehicles.

The density of traffic is then defined as: ρ=N.

The traffic conditions can be classified by the system into: Low Density, Medium Density, High Density. It does not use complicated tracking systems but uses live counts of detecting them, which allows it to be processed much faster and be implemented with ease.

E. Traffic Logic Module (Dynamic Signal Optimization)

A traffic control algorithm is applied which is a rule based one and adjusts the signal timing dynamically depending on the perceived vehicle density. The duration of the green signal Gis was obtained as: G=k.N. where: N= number of observed vehicles, k= scaling factor. Green lights are given longer to lanes that contain a greater number of vehicles and shorter to the lanes that do not have a lot of traffic. This dynamic mechanism enhances the traffic flow as opposed to the fixed-timer systems.

F. Ambulance Priority Module

The system is equipped with an emergency management system of ambulances. In the case of the detection of an ambulance by YOLOv11, the regular traffic signal system is inhibited. The adjacent lane is then provided with green light to pass through without any further interruption. This saves the time within which the emergency response is provided and improves the efficiency of the traffic management system.

FIG.3: Ambulance Detection

G. Dashboard and Analytics Module

The visualization of system outputs is in real time and is done in a Flask based web dashboard. The dashboard will show: Live video feed with detected overlay, Vehicle count and density level of traffic, Signal timing status, Simple traffic analytics. This module is more of a decision-support system, which lets users track the traffic and manage it effectively.

FIG.4: Dashboard Module

H. Traffic Analysis and Reporting Module

The system captures traffic information and creates insights depending on the traffic trends. This is: Vehicle counts trends, level of congestion analysis, Signal timing effectiveness. These reports will also be used to assess the performance of the system, and may assist in planning and optimization of future traffic.

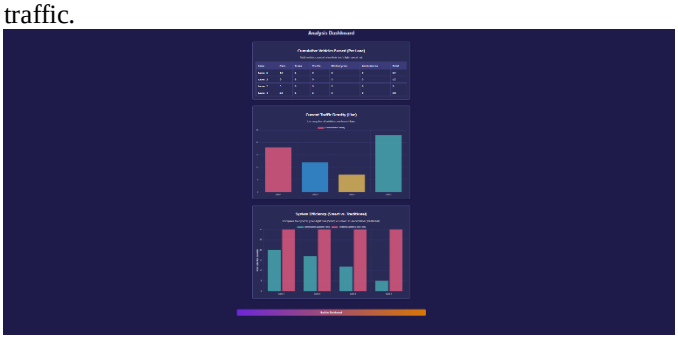


FIG.5: Traffic Analysis Module

I. User Authentication Module

This is the first aspect of the module. The system also has a rudimentary authentication system that has been used to access it securely. Role-based access control (admin/user) gives users a chance to register and log in. This guarantees that user operations in the system like in monitoring, analytics and control are accessed by authorized users.

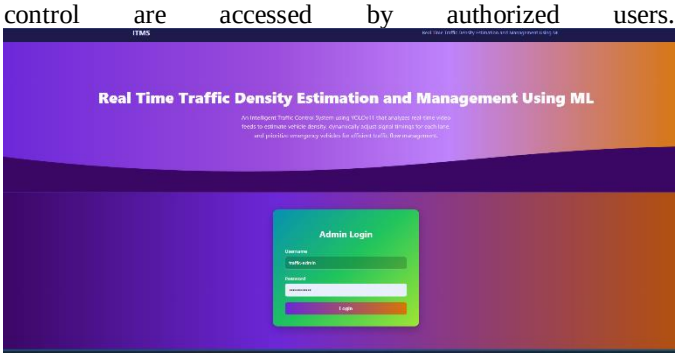


FIG.6: User Authentication Module

IV. IMPLEMENTATION

To develop the proposed system, the Python language is used to incorporate computer vision, deep learning and web-based technologies to create a fully real-time traffic density estimation and management pipeline. It is implemented with the help of libraries like OpenCV to process the video, NumPy to perform numeric calculations, and Flask to create the web-based dashboard interface, and also YOLOv11 is taken as the baseline deep learning model to detect objects. The system will be able to take any type of input, be it a live CCTV camera feed or pre-recorded traffic videos, so that it will be flexible in deployment during live-time as well as during offline testing. OpenCV is used to process the input video stream continuously, and it is decomposed into single frames with an approximate rate of 2030 frames per second so that real-time analysis of the traffic conditions can be carried out without much latency.

Each of the extracted frames is subjected to a sequence of preprocessing steps that are meant to standardize the input frame to enhance the performance of the detection. The frames are then resized to a fixed resolution (e.g. 640×640 pixels) to match the input requirements of the YOLOv11 model and to guarantee the processing speed to be similar. Noise mitigation measures like Gaussian filtering are used in order to reduce the impact of environmental interference, like shadows, glare and weather change. The normalization of pixels is carried out in order to ensure the same intensity distribution across the frames and this stabilizes the model predictions. Moreover, there is a defined Region of Interest (ROI) that is used to isolate the area of interest like the lanes or even the junction regions to minimize the background information and increase the efficiency of the computation.

The processed frames are then fed to the YOLOv11 object detection network which does real time inference to identify vehicles and ambulances. The model makes use of single-stage detection architecture which facilitates simultaneous localization and classification of objects on a frame. YOLOv11 produces the bounding boxes, label of the classes (car, bus, truck, motorcycle, and ambulance) and confidence scores of each detection in every frame. A confidence level (such as 0.5) is used to remove low-confidence detections and minimize false positives. The model especially works well under dense traffic scenarios having several vehicles visible together and is also able to recognize small or remote objects owing to the enhanced feature extraction aptitude.

The system instead of using computationally intensive tracking algorithms uses frame-based vehicle counting approach where the detected vehicle count in each frame is directly computed. This will make the overall architecture simpler and process quicker, which is appropriate to the real-time application. The identified number of vehicles is then employed to estimate the density of traffic in the individual lane or area. Traffic density is calculated as a value of the number of vehicles in the considered ROI and the system determines the density into discrete levels of low, medium, and high depending on

predefined threshold levels. They are set empirically with reference to the common traffic conditions in the dataset. The estimated traffic density is then forwarded to a rule-based traffic signal control module which dynamically adjusts the signal timing. The duration of the green signal is computed in proportion to the number of vehicles that have been detected such that lanes that are congested are allowed to have a longer green signal as compared to those that are less congested and are given shorter green signal. This dynamic adjustment enhances the efficiency of traffic flow and minimizes unjustified waiting time in comparison to the conventional systems that had fixed timers. The system also has the ambulance priority system, where any ambulance passing through the camera is detected by YOLOv11, and signal logic is overridden immediately. In these situations, the lone that will automatically be signaled a green light will permit the ambulance to go through and thus emergency response is enhanced.

The last step in the implementation is the stage of visualization and interaction with a user by a Flask-based web dashboard. The dashboard shows the live video feed with bounding boxes drawn around the identified vehicles, with real-time data displayed in the form of the number of vehicles present in traffic, the levels of the traffic density, and the status of the signals. This gives a clear and easy-to-use interface on how to monitor the traffic and the system behavior. It is also combined with an SQLite database that is used to authenticate users enabling secure access of the system users and control access based on roles. The database is a database of user credentials and has simple functions of registration and login. Altogether, the implementation is a data acquisition, preprocessing, detection, analysis, decision-making, and visualization pipeline that combines all these components into a unified system that is capable of managing traffic in real-time and in an intelligent way.

V. RESULTS AND PERFORMANCE ANALYSIS

The proposed system was tested on the accessibility of real-time CCTV video feeds and publicly available traffic data under various traffic conditions, such as low, medium, and high-density conditions. Detection accuracy, processing speed, and success of the traffic density estimation and signal control are analyzed as the performance of the system.

TABLE 1: Vehicle Detection Accuracy (YOLOv11)

Vehicle Type	Detection Model	Accuracy
Car	YOLOv11	High (~97%)
Bus	YOLOv11	High (~96%)
Truck	YOLOv11	High (~95%)
Motor Bike	YOLOv11	Moderate (~92%)
Ambulance	YOLOv11	High (~95%)

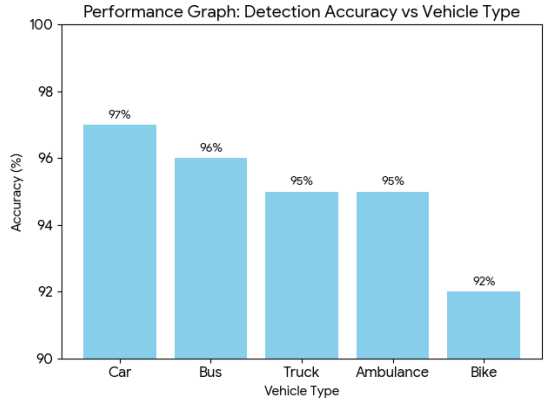
TABLE 2: Traffic Density Estimation Analysis

Situation	Vehicle Count	Density Level	Signal Response
Situation 1	High (30+)	High	Long Green Time
Situation 2	Medium (15–25)	Medium	Moderate Green Time
Situation 3	Low (<10)	Low	Short Green Time
Situation 4	High	High	Priority Allocation
Situation 5	Medium	Medium	Balanced Timing
Situation 6	Low	Low	Minimal Delay
Situation 7	Mixed	Medium	Adaptive Control

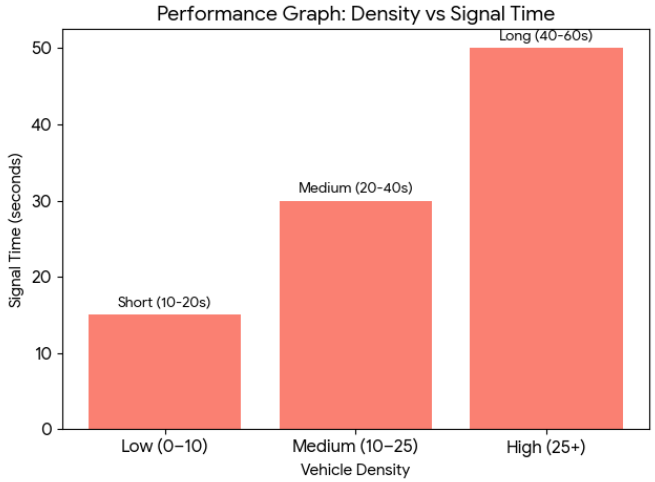
TABLE 3: System Performance Metrics

Parameter	**Value**
Detection Model	YOLOv11
Average Accuracy	~95–97%
Processing Speed	20–30 FPS
Input Type	CCTV Video
Response Time	Real-time
Platform	Python + OpenCV + Flask

Performance Graph 1: Detection Accuracy vs Vehicle Type



Performance Graph 2: Density vs Signal Time



Analysis of Results

The experiment and its findings show that the suggested system has great detection rates in various categories of vehicles, and YOLOv11 can detect a vehicle even in a severe and filled traffic situation. This provides real-time performance and averages 20-30 frames per second which makes the system appropriate in the live traffic monitoring application. The density-based signal control system is able to adjust to the changing traffic conditions with ease as it provides the correct amount of green signal time, which minimizes the number of congestions and maximizes the efficiency of traffic flow. Also, the system demonstrates robust functionality without the need of sophisticated tracking algorithms, it minimizes the computational burden, and yet it is practical in its accuracy. The ambulance detection feature also introduces the system to a higher level by allowing signal control in the case of an emergency; a priority must be given to the ambulance. Altogether, the findings confirm the fact that the suggested solution is efficient, scalable, and real-time to apply in intelligent traffic management.

VI. COMPARATIVE STUDY

The traffic management strategies persistent in the available literature are largely structured in a similar framework, which comprises of data collection, extraction of traffic related parameters, number of vehicles and flow rate, estimation of traffic density, and application of traffic signal controller. Nonetheless, most legacy systems use attached signals or infrastructure based on sensors, which are not flexible to respond to real-time road situations. Recent intelligent traffic systems in the urban setting, including AI-driven signal coordination and IoT-based traffic monitoring have demonstrated better results in traffic management, yet they typically demand multifaceted infrastructure, high implementation costs, and combination with a variety of other systems.

Contrary to that, the recommended system is vision-based and works with CCTV video feeds and computer vision algorithms, which do not require any other physical sensors. The system employs a video processing system based on OpenCV and an object detection system based on YOLOv11 that is capable of identifying vehicles accurately in different traffic situations. Contrary to the previous approaches that used classical image processing algorithms like edge detection or background subtraction, the given approach uses deep learning to detect vehicles in a robust and scalable way.

Moreover, the classical methods, especially those based on the comparison of reference images or the complicated tracking techniques to count vehicles, may be subject to errors and may raise the level of computational complexity. Compared to it, the proposed system applies a frame-based detection and counting approach which simplifies the pipeline and preserves a high-quality traffic density estimate. The calculated number of vehicles is directly transformed into calculating the level of traffic density that is then applied to a rule-based signal control algorithm to manage traffic dynamically. On the whole, the offered solution is more convenient and cost-efficient, and it covers real-time object detection with adaptive signal management. It lowers reliance on hardware infrastructure, provides a higher degree of scalability, and it is more adaptable to real-world traffic patterns, which is why it is compatible with implementation in a modern smart city setting.

VII. CONCLUSION

This work presents a real-time traffic density estimation and This paper introduces a real-time traffic congestion estimation and control system which is dynamic to overcome the constraints of the traditional traffic control schemes that operate on fixed timers and manual control. The proposed system is able to detect cars using CCTV video streams and deploying the YOLOv11 object detecting model, which can properly identify traffic vehicles in various traffic conditions and determine the number of vehicles in the road. Vision-based approach avoids the use of costly sensor-based infrastructure; thus, the system is more scalable and affordable to be implemented in actual practice in urban settings. The application proves that real-time vehicle detection and density-based analysis can be useful in optimizing the dynamic traffic signal. The control mechanism which is based on rules adjusts the timing of the signals to the conditions of the traffic whereby the lanes that are highly congested are allocated more time to green signals as compared to those that are less congested so that the time wastage is eliminated by clearing the lanes effectively. This adaptive technology saves wastage of time at the parking areas, increases traffic flow and boosts the overall use of the roads. Moreover, the system is real time performance and well-processed, and therefore, it can be applied in the real-time monitoring of continuous performance in live traffic conditions.

Another advantage of the system is the ambulance detection and prioritized signal control that ensures that emergency vehicles move ahead through intersections without wasting time. This aspect emphasizes the practicality of the system in

enhancing the response time in cases of emergencies and safety of the people. Its integration of a Flask-based dashboard is available to visualize traffic conditions, vehicle detection findings, and signal state in real-time, which allows one to successfully monitor and make quality decisions.

In general, the suggested system provides a straightforward but efficient approach to intelligent traffic control, which involves real-time-based detection, density estimation, and adaptive signal control within a single pipeline. Although the existing system is concentrated on the detection-based density estimation without intricate tracking and predictive modelling, it offers a solid background to the developments in the future. Possible changes would be the addition of multi-cameras to monitor the traffic in a large scale, the ability to predict the traffic congestion with the help of the predictive models, and the possibility to expand the system to detect the traffic violations and integrate the smart city infrastructure. It can also be improved with further developments; hence, it makes the system a strong and complete solution to the next-generation intelligent transportation systems.

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